

# **GEN AI DESIGNATHON**

**Project Name:** Enterprise RAG Knowledge  
Assistant with Document Intelligence

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## INDEX

| <b>S.I NO</b> | <b>CONTENT</b>                                       | <b>PAGE NO</b> |
|---------------|------------------------------------------------------|----------------|
| <b>1</b>      | <b>EXECUTIVE SUMMARY</b>                             | <b>3</b>       |
| <b>2</b>      | <b>PROJECT OVERVIEW</b>                              | <b>5</b>       |
| <b>3</b>      | <b>BUSINESS PROBLEM</b>                              | <b>7</b>       |
| <b>4</b>      | <b>TECHNOLOGY STACK</b>                              | <b>8</b>       |
| <b>5</b>      | <b>HIGH LEVEL ARCHITECTURE</b>                       | <b>11</b>      |
| <b>6</b>      | <b>COMPLETE LLM WORKFLOW</b>                         | <b>13</b>      |
| <b>7</b>      | <b>PROJECT FOLDER STRUCTURE</b>                      | <b>15</b>      |
| <b>8</b>      | <b>FRONTEND ARCHITECTURE</b>                         | <b>17</b>      |
| <b>9</b>      | <b>BACKEND ARCHITECTURE</b>                          | <b>18</b>      |
| <b>10</b>     | <b>AZURE &amp; OPEN AI SERVICES<br/>INTEGRATIONS</b> | <b>20</b>      |
| <b>11</b>     | <b>SCREENSHOTS</b>                                   | <b>21</b>      |
| <b>12</b>     | <b>CONCLUSION</b>                                    | <b>23</b>      |

## **1. Executive Summary**

### **1.1 Introduction**

The Enterprise Retrieval-Augmented Generation (Enterprise RAG) Platform is an AI-powered enterprise knowledge management solution designed to enable employees to retrieve accurate, secure, and cited information from organizational documents through natural language conversations.

Traditional enterprise search systems rely on keyword matching, making it difficult for employees to locate relevant information across large document repositories. The Enterprise RAG Platform addresses this challenge by combining semantic search, Retrieval-Augmented Generation (RAG), Azure OpenAI GPT models, and Role-Based Access Control (RBAC) into a unified solution.

The platform retrieves relevant enterprise documents, validates user authorization, generates responses using Azure OpenAI, performs guardrail validation, and returns responses with citations while maintaining complete audit logs.

### **1.2 Business Objectives**

The primary objectives of the Enterprise RAG Platform are:

- Improve enterprise knowledge discovery using semantic search.
- Reduce employee search time by providing conversational access to enterprise documents.
- Generate accurate, context-aware responses with document citations.
- Protect sensitive information through Role-Based Access Control (RBAC).
- Ensure compliance using audit logging and response validation.
- Provide a scalable architecture suitable for enterprise deployment.

### **1.3 Key Features**

The platform includes the following major capabilities:

- AI-powered conversational interface.
- Retrieval-Augmented Generation (RAG).
- Azure OpenAI GPT integration.
- Semantic document retrieval.
- Enterprise document indexing.
- Role-Based Access Control (RBAC).
- Guardrail-based response validation.
- Human handoff workflow.
- Audit logging.
- Azure Blob Storage integration.
- FastAPI-based backend services.
- Modern web-based frontend.

## **1.4 Business Benefits**

The Enterprise RAG Platform delivers several benefits to organizations:

- Faster access to enterprise knowledge.
- Improved decision-making through accurate responses.
- Reduced dependency on manual document searches.
- Enhanced security through RBAC.
- Increased trust through source citations.
- Better compliance through audit trails.
- Reduced operational costs.
- Improved employee productivity.

## **1.5 Project Scope**

The project includes:

- Enterprise document ingestion.
- Semantic search.
- LLM-powered answer generation.
- User authorization.
- Guardrail validation.
- Audit logging.
- Frontend chat application.
- Backend API services.
- Azure OpenAI integration.
- Azure Blob Storage integration.

The project excludes:

- Model training.
- Custom LLM development.
- Enterprise Identity Provider implementation.
- OCR for scanned documents.
- Multi-language translation.

## **2. Project Overview**

### **2.1 Introduction**

Enterprise organizations generate and maintain thousands of documents, including policies, procedures, reports, manuals, and technical documentation. Employees often spend significant time searching across multiple repositories to locate relevant information.

The Enterprise RAG Platform provides a centralized conversational interface that enables users to ask questions in natural language. The system retrieves relevant information from enterprise documents, validates user permissions, generates AI-powered responses, and returns answers supported by citations.

### **2.2 Existing System**

Traditional enterprise knowledge management systems typically rely on:

- Keyword-based document search.
- Manual browsing.
- Static document repositories.
- Limited access control.
- No AI-assisted summarization.
- No contextual understanding.
- Minimal response validation.

These limitations often result in increased search time, inconsistent information retrieval, and reduced productivity.

### **2.3 Proposed System**

The proposed Enterprise RAG Platform enhances knowledge retrieval by integrating semantic search with Azure OpenAI GPT models.

The system processes user queries through a structured workflow:

1. Accept user query.
2. Retrieve relevant document chunks.
3. Build contextual prompts.
4. Generate AI response.
5. Validate response using guardrails.
6. Apply retry logic if necessary.
7. Escalate to human handoff when required.
8. Return formatted response with citations.

## 2.4 System Users

The platform supports multiple user roles.

| User Role     | Responsibilities                                         |
|---------------|----------------------------------------------------------|
| Administrator | Manage users, documents, roles, and system configuration |
| Manager       | Access department-specific enterprise knowledge          |
| Employee      | Query authorized enterprise documents                    |
| Guest         | Access public documents only                             |

## 2.5 Core Functionalities

The system provides the following capabilities:

### Document Management

- Upload enterprise documents.
- Store documents securely.
- Maintain document metadata.
- Manage document versions.

### Knowledge Retrieval

- Semantic search.
- Similarity matching.
- Top-K retrieval.
- Context generation.

### AI Response Generation

- Prompt engineering.
- GPT response generation.
- Citation generation.
- Response formatting.

### Security

- User authentication.
- RBAC authorization.
- Audit logging.
- Access validation.

### Response Validation

- Guardrail evaluation.
- Citation verification.
- Retry workflow.
- Human handoff.

## 2.6 Expected Outcomes

The implementation of the Enterprise RAG Platform is expected to:

- Improve search efficiency.
- Increase response accuracy.
- Reduce manual effort.
- Improve knowledge accessibility.
- Enhance enterprise security.
- Improve compliance.
- Support scalable enterprise deployment.

## 3. Business Problem

### 3.1 Background

Organizations store information in multiple places such as PDFs, emails, manuals, and knowledge bases. This makes finding the right information difficult and time-consuming.

### 3.2 Existing Challenges

- **Scattered Information:** Documents are stored in different systems.
- **Keyword Search:** Cannot understand user intent.
- **Slow Search:** Employees spend too much time searching.
- **Low Reliability:** Sources are difficult to verify.
- **Security Risks:** Unauthorized access to sensitive data.
- **Compliance Issues:** Limited audit tracking.
- 

### 3.3 Business Impact

- Lower productivity
- Higher operational costs
- Slower decisions
- Duplicate knowledge
- Security risks
- Compliance issues

### 3.4 Proposed Solution

The **Enterprise RAG Platform** uses **Azure OpenAI** and **semantic search** to provide:

- Natural language search
- AI-powered answers
- Secure document access
- Citation-based responses

- Audit logging
- Human escalation

### 3.5 Expected Benefits

| Objective     | Benefit             |
|---------------|---------------------|
| Faster Search | Saves time          |
| AI Responses  | Higher productivity |
| RBAC Security | Secure data access  |
| Citations     | Trusted answers     |
| Audit Logs    | Compliance support  |
| Azure Cloud   | Scalable solution   |
| Human Handoff | Better support      |

## 4. Technology Stack

The Enterprise RAG Platform is built using modern cloud-native technologies that enable scalable document processing, secure AI-powered knowledge retrieval, and enterprise-grade governance. Each technology has been selected to provide high performance, flexibility, and seamless integration with Azure services.

### 4.1 Technology Overview

| Layer               | Technology                          | Purpose                                                |
|---------------------|-------------------------------------|--------------------------------------------------------|
| Frontend            | HTML5, CSS3, JavaScript             | Provides an interactive web-based user interface       |
| Backend             | Python, FastAPI                     | REST API development and business logic                |
| AI Model            | Azure OpenAI GPT-4                  | Generates context-aware responses                      |
| Embedding Model     | text-embedding-3-large              | Converts document chunks into vector embeddings        |
| Knowledge Retrieval | TF-IDF / Azure AI Search            | Retrieves relevant documents using semantic similarity |
| Storage             | Azure Blob Storage                  | Stores enterprise documents securely                   |
| Database            | SQLite                              | Stores audit logs and application metadata             |
| Authentication      | Role-Based Access Control (RBAC)    | Controls user access based on roles                    |
| Monitoring          | Azure Monitor, Application Insights | Tracks system health and application performance       |

## 4.2 Frontend Technologies

The frontend provides an intuitive chat interface for users to interact with the Enterprise RAG Platform.

### Technologies Used

- HTML5
- CSS3
- JavaScript

### Responsibilities

- User Login
- Chat Interface
- Document Upload
- Display AI Responses
- Display Citations
- Error Notifications
- Response History

## 4.3 Backend Technologies

The backend manages API requests, document retrieval, authorization, AI orchestration, and response validation.

### Technologies Used

- Python
- FastAPI

### Responsibilities

- Process user requests
- Handle document uploads
- Manage authentication
- Execute retrieval pipeline
- Integrate Azure OpenAI
- Apply RBAC
- Validate responses
- Maintain audit logs

## 4.4 Azure OpenAI

Azure OpenAI provides Large Language Model capabilities for generating intelligent responses.

### Model Used

- GPT-4

### Responsibilities

- Context-aware response generation
- Prompt processing

- Response summarization
- Citation-aware answer generation

#### **4.5 Embedding Model**

The embedding model converts enterprise documents into numerical vectors.

##### **Model**

- text-embedding-3-large

##### **Purpose**

- Semantic similarity search
- Context retrieval
- Knowledge indexing

#### **4.6 Retrieval Engine**

The retrieval engine identifies the most relevant documents based on the user's query.

##### **Technologies**

- TF-IDF
- Azure AI Search (Production)

##### **Responsibilities**

- Document indexing
- Similarity search
- Ranking
- Top-K retrieval

#### **4.7 Azure Blob Storage**

Azure Blob Storage serves as the centralized repository for enterprise documents.

##### **Stores**

- PDF Files
- Word Documents
- Text Files
- Metadata
- Uploaded Documents

##### **Benefits**

- High availability
- Scalability
- Secure storage
- Easy integration

## 4.8 SQLite Database

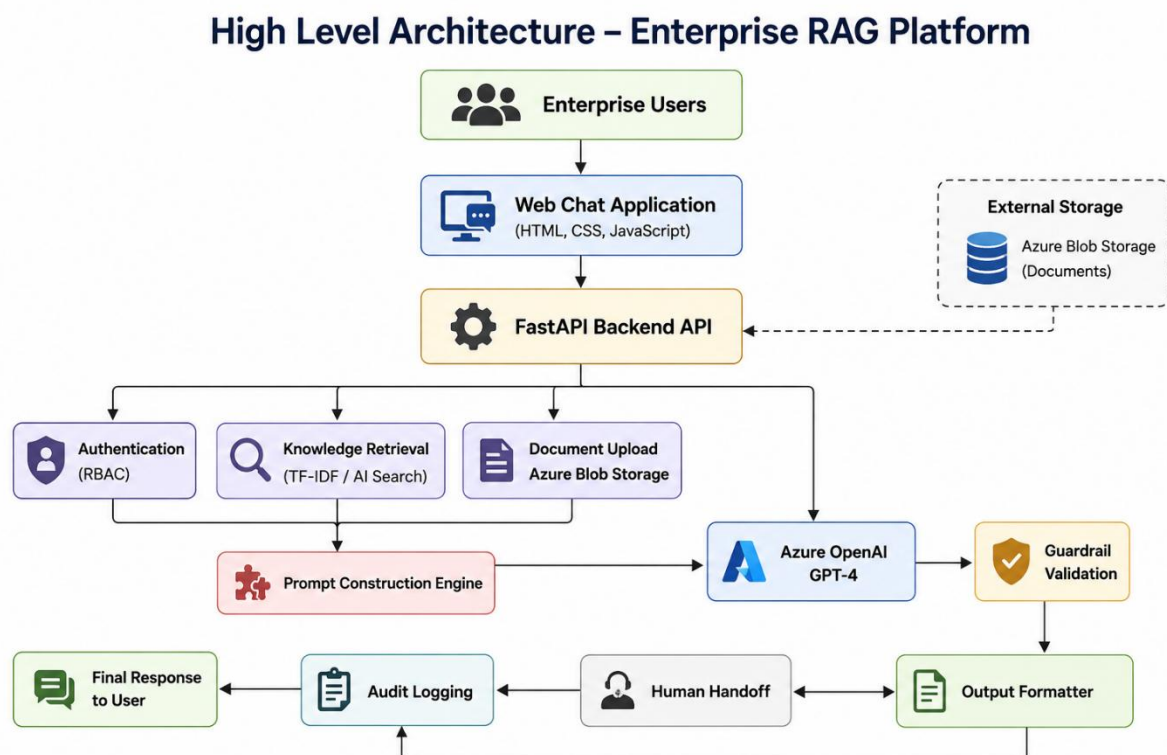
SQLite is used to maintain audit logs and application metadata.

### Stores

- User Queries
- Generated Responses
- Citations
- Access Logs
- Error Logs

## 5. High-Level Architecture

The Enterprise RAG Platform follows a modular architecture that separates user interaction, business logic, knowledge retrieval, AI processing, validation, and storage into independent layers. This modular design improves scalability, maintainability, and security.



## 5.1 Component Architecture

The application consists of six major components:

### 1. Presentation Layer

#### Responsibilities:

- User Login
- Chat Interface

- Document Upload
- Response Display

## **2. API Layer**

### **Responsibilities:**

- Handle API requests
- Validate requests
- Route requests
- Return responses

## **3. Retrieval Layer**

### **Responsibilities:**

- Search indexed documents
- Rank relevant chunks
- Retrieve Top-K results
- Build retrieval context

## **4. AI Layer**

### **Responsibilities:**

- Create prompts
- Process with GPT
- Generate responses
- Generate citations

## **5. Validation Layer**

### **Responsibilities:**

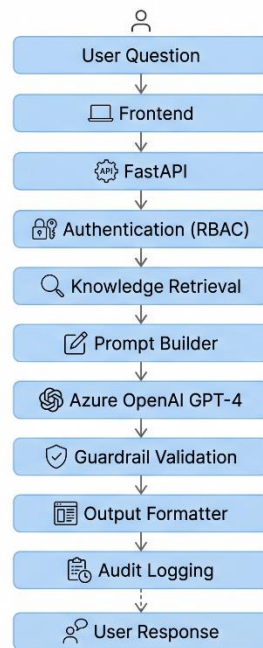
- Validate citations
- Enforce policies
- Detect hallucinations
- Retry if needed

## **6. Storage Layer**

### **Responsibilities:**

- Store documents
- Store metadata
- Maintain audit logs
- Maintain user logs

## 5.2 End-to-End Data Flow



## 6. Complete LLM Workflow

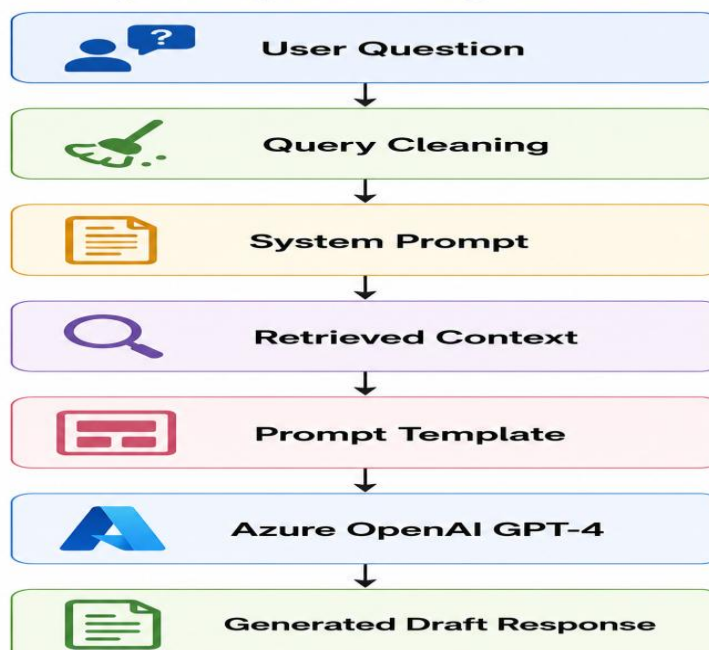
The LLM workflow is the core intelligence of the Enterprise RAG Platform. It orchestrates document retrieval, prompt construction, response generation, validation, and formatting to ensure users receive accurate, secure, and cited answers.

### 6.1 Prompt Engineering Workflow

#### Objective

Construct a high-quality prompt by combining user input with retrieved enterprise knowledge before sending it to Azure OpenAI.

#### Prompt Engineering Workflow



## Components

- User Query
- System Prompt
- Retrieved Context
- Prompt Template
- GPT-4 Model

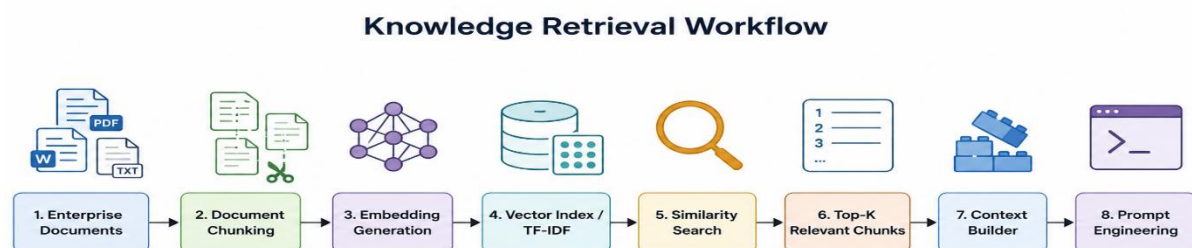
## Output

A context-aware draft response grounded in enterprise documents.

## 6.2 Knowledge Retrieval Workflow

### Objective

Retrieve the most relevant document chunks using semantic similarity before response generation



### Key Features

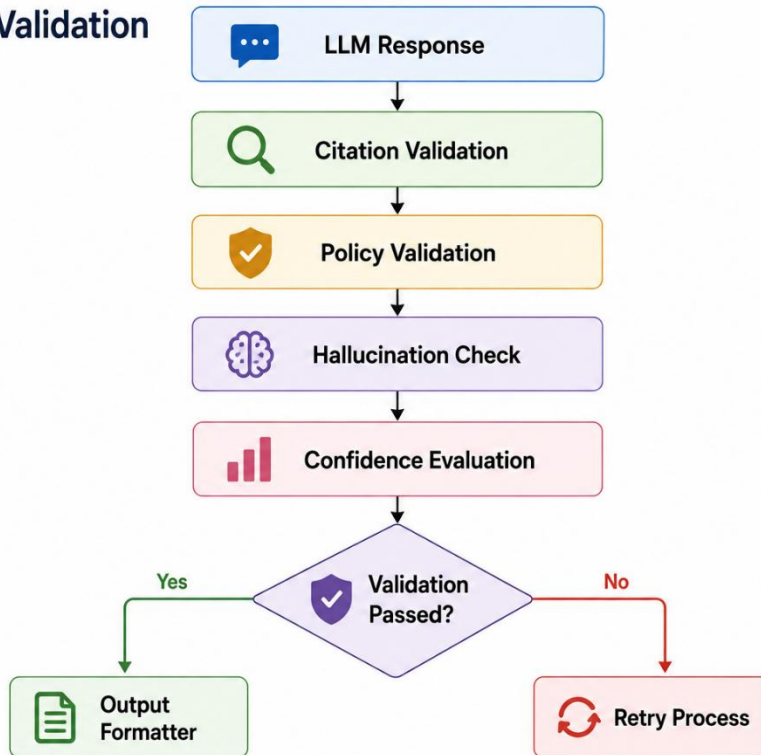
- Document chunking
- Embedding generation
- Similarity search
- Top-K retrieval
- Context preparation

## 6.3 Guardrail & Validation Workflow

### Objective

Validate AI-generated responses for accuracy, compliance, and citation integrity before returning them to the user.

## Guardrail & Validation Workflow



### Validation Checks

- Citation completeness
- Response relevance
- Policy compliance
- Hallucination detection
- Confidence threshold

## 7. Project Folder Structure

The Enterprise RAG Platform follows a modular and layered architecture to improve maintainability, scalability, and code reusability. Each module is responsible for a specific function, making the application easier to develop, test, and deploy.

Project Directory Structure

enterprise-rag-platform/

```
|
|
|— frontend/
|   |— index.html
|   |— styles.css
|   |— app.js
|   └— assets/
|
|— backend/
|   |— main.py
|   |— pipeline.py
|   └— retrieval.py
```

```

| |— access_control.py
| |— llm_client.py
| |— guardrail.py
| |— storage.py
| |— audit.py
| |— config.py
| |— utils.py
| |
| — requirements.txt
| — .env
| — Dockerfile
| — README.md
| — docker-compose.yml

```

## 7.1 Project Directory Structure

| Module            | Description                   |
|-------------------|-------------------------------|
| frontend          | User Interface                |
| backend           | FastAPI Application           |
| retrieval.py      | Retrieves relevant documents  |
| pipeline.py       | Controls end-to-end workflow  |
| llm_client.py     | Azure OpenAI integration      |
| access_control.py | Role-Based Access Control     |
| guardrail.py      | AI response validation        |
| storage.py        | Azure Blob Storage operations |
| audit.py          | Audit logging                 |
| config.py         | Configuration management      |
| documents         | Enterprise knowledge base     |
| uploads           | Uploaded files                |
| database          | SQLite audit database         |
| logs              | Application logs              |

## 8. Frontend Architecture

### 8.1 Overview

The frontend provides a responsive web interface that allows users to interact with the Enterprise RAG Platform through a conversational chat interface.

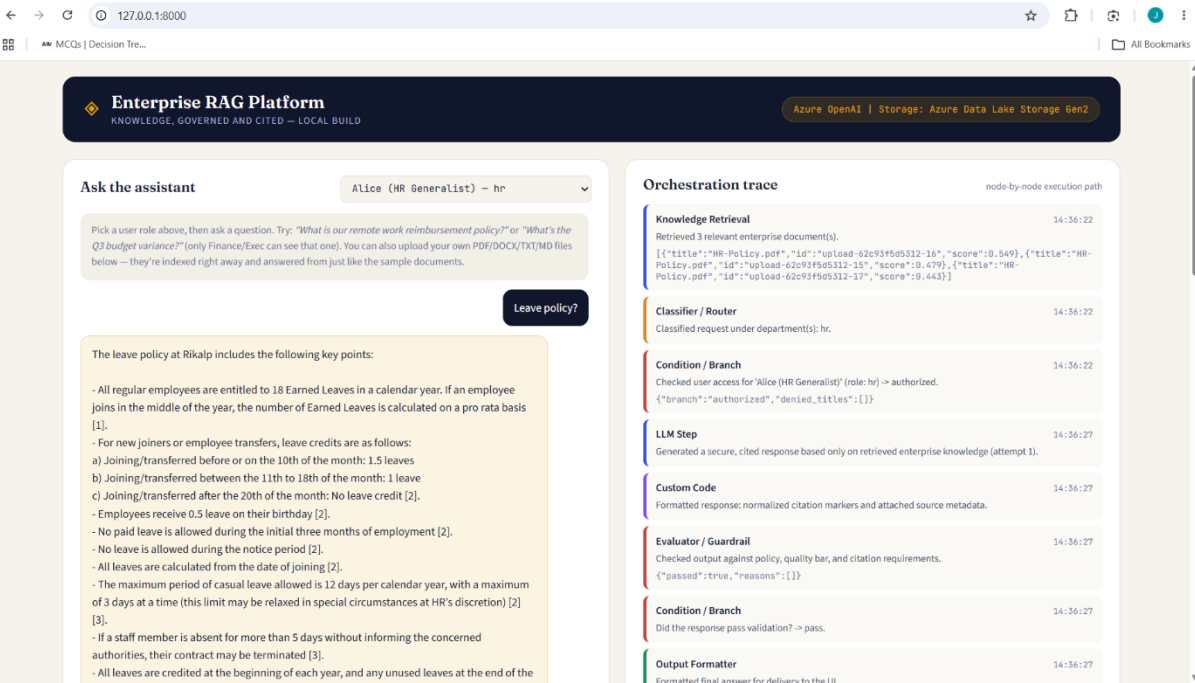
The frontend communicates with the FastAPI backend using REST APIs.

### 8.2 Frontend Components

Frontend UI

| Component        | Purpose                              |
|------------------|--------------------------------------|
| Login            | User authentication                  |
| Chat Window      | User interaction with AI             |
| Upload Documents | Upload enterprise documents          |
| Response Panel   | Display AI-generated responses       |
| Citation Panel   | Show source references               |
| Error Messages   | Display validation and system errors |

Screenshots

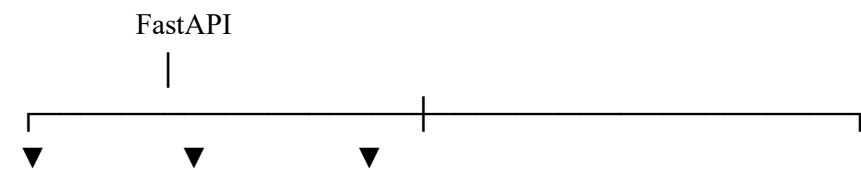


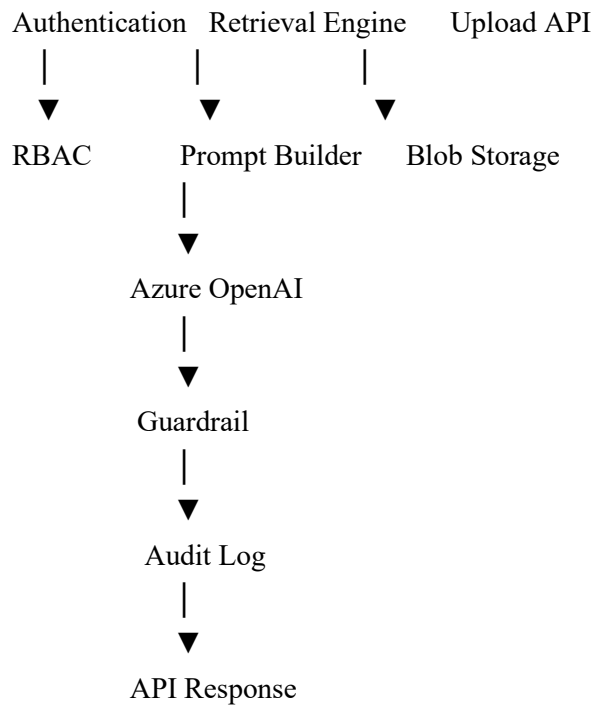
9. Backend Architecture

9.1 Overview

The backend is implemented using FastAPI and serves as the central orchestration layer. It processes incoming requests, retrieves relevant knowledge, invokes Azure OpenAI, validates responses, and returns formatted results.

9.2 Backend Component Diagram





### 9.3 Backend Modules

#### **main.py**

- Application entry point
- API routing
- FastAPI initialization

#### **pipeline.py**

Responsible for:

- Complete request orchestration
- Calling retrieval module
- Calling GPT
- Guardrail validation
- Output formatting

#### **retrieval.py**

Responsible for:

- Semantic search
- Document retrieval
- Ranking
- Top-K selection

#### **llm\_client.py**

Responsible for:

- Azure OpenAI communication
- Prompt submission

- Response generation

#### **access\_control.py**

Responsible for:

- User authorization
- Role validation
- Permission checking

#### **guardrail.py**

Responsible for:

- Hallucination detection
- Citation validation
- Policy validation
- Retry workflow

#### **storage.py**

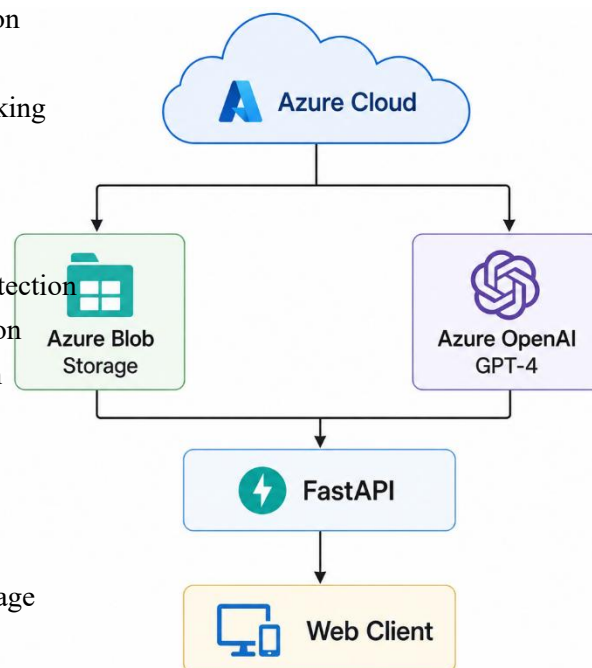
Responsible for:

- Azure Blob Storage
- Upload
- Download
- Metadata

#### **audit.py**

Responsible for:

- Audit logging
- Query history
- Error tracking
- User activity



## **10. Azure & OpenAI Services Integration**

### **10.1 Overview**

The Enterprise RAG Platform integrates Azure cloud services with Azure OpenAI to provide secure, scalable, and AI-powered enterprise knowledge retrieval.

## 10.2 Azure Blob Storage

### Purpose

Stores enterprise documents securely.

### Operations

- Upload Documents
- Download Documents
- Metadata Storage
- File Versioning

## 10.3 Azure OpenAI

### Models

| Model                  | Purpose              |
|------------------------|----------------------|
| GPT-4                  | Answer Generation    |
| text-embedding-3-large | Embedding Generation |

## 11.Screenshots:

127.0.0.1:8000

File upload and doc...

Enterprise RAG Platform

KNOWLEDGE, GOVERNED AND CITED — LOCAL BUILD

Azure OpenAI | Storage: Azure Data Lake Storage Gen2

Ask the assistant

Alice (HR Generalist) — hr

Pick a user role above, then ask a question. Try: "What is our remote work reimbursement policy?" or "What's the Q3 budget variance?" (only Finance/Exec can see that one). You can also upload your own PDF/DOCX/TXT/MD files below — they're indexed right away and answered from just like the sample documents.

what are the skills in the resume?

The skills listed in the resume are:

- Languages: Java, Python (Intermediate)

- Tools: Excel, PowerPoint, MySQL

- Dev Tools: Git, GitHub

- Soft Skills: Communication, Teamwork [1]

[1] JAYAKUMAR\_RESUME.pdf

[2] JAYAKUMAR\_RESUME.pdf

[3] JAYAKUMAR\_RESUME.pdf

Ask a question about company knowledge...

Ask

Orchestration trace

node-by-node execution path

Knowledge Retrieval

08:51:34

Retrieved 3 relevant enterprise document(s).

[{"title": "JAYAKUMAR\_RESUME.pdf", "id": "upload-85f57544d8bf-1", "score": 0.392}, {"title": "JAYAKUMAR\_RESUME.pdf", "id": "upload-85f57544d8bf-3", "score": 0.276}, {"title": "JAYAKUMAR\_RESUME.pdf", "id": "upload-85f57544d8bf-2", "score": 0.246}]

Classifier / Router

08:51:34

Classified request under department(s): hr.

Condition / Branch

08:51:34

Checked user access for 'Alice (HR Generalist)' (role: hr) -> authorized.

{"branch": "authorized", "denied\_titles": []}

LLM Step

08:51:37

Generated a secure, cited response based only on retrieved enterprise knowledge (attempt 1).

Custom Code

08:51:37

Formatted response: normalized citation markers and attached source metadata.

Evaluator / Guardrail

08:51:37

Checked output against policy, quality bar, and citation requirements.

{"passed": true, "reasons": []}

Condition / Branch

08:51:37

Did the response pass validation? -> pass.

Output Formatter

08:51:37

Formatted final answer for delivery to the UI.

35°C

Partly sunny

Search

ENG

IN

14:21

02-08-2026

## DASHBOARD

| + Deploy model                                    |                        |               |           |                  | Columns |  |
|---------------------------------------------------|------------------------|---------------|-----------|------------------|---------|--|
| Refresh Edit Delete Open in playground Reset view |                        |               |           |                  |         |  |
| Name                                              | Model name             | Model version | State     | Model retirement |         |  |
| <a href="#">gpt-4.1</a>                           | gpt-4.1                | 2025-04-14    | Succeeded | Apr 14, 2027 5:  |         |  |
| <a href="#">text-embedding-3-large</a>            | text-embedding-3-large | 1             | Succeeded | Feb 9, 2028 5:3  |         |  |

## OPEN AI DEPLOYMENT MODEL

21 | Page

File upload and doc...

Microsoft Foundry

New Foundry

All resources (1)

Create project

Search

Type

Subscription

Region

| Name          | Type         | Parent resource | Subscription    | Region  | Created by                 |
|---------------|--------------|-----------------|-----------------|---------|----------------------------|
| openaiprocess | Azure OpenAI | --              | Learner Account | East US | azuser7236_mmml.local@kart |

1-1 of 1

openaiprocess

Azure OpenAI

Last modified: 8/2/2026, 2:01:15 PM

Open in Foundry Classic

Resource details

Type

Azure OpenAI

Location

East US

Public network access

Enabled

Resource group

rg-azuser7236\_mmml.local-fyvqk

Resource group ID

/subscriptions/4116346b-5ded-4f3b-8387-f8d055802adc/

Subscription

Learner Account

Subscription ID

4116346b-5ded-4f3b-8387-f8d055802adc

View resource

Manage this resource in Azure Portal

## AZURE OPEN AI DEPLOYMENT MODEL

## 12. Conclusion

The Enterprise RAG Platform is an AI-powered enterprise knowledge management solution designed to provide secure, intelligent, and context-aware access to organizational knowledge. By combining Retrieval-Augmented Generation (RAG), Azure OpenAI GPT-4, semantic document retrieval, and Role-Based Access Control (RBAC), the platform enables users to retrieve accurate, cited, and authorized information through a conversational interface.

The solution integrates document ingestion, semantic retrieval, prompt engineering, response validation, and audit logging into a single workflow. Every user request is authenticated, authorized, validated, and monitored before the final response is delivered.

The modular architecture allows organizations to scale the platform while maintaining security, governance, and compliance requirements.